Skills Summary

Scale Factor Confusion: Linear, Square, Cubic

Use this reference while completing your worksheet or when studying.

Skill 1 — Scaling an area: multiply by the square of k

Rule: for similar figures with linear scale factor k,

$$\frac{\text{new length}}{\text{old length}} = k, \qquad \frac{\text{new area}}{\text{old area}} = k^2, \qquad \frac{\text{new volume}}{\text{old volume}} = k^3$$

Why: an area is built from two perpendicular lengths and a dilation stretches both, so the factor is used twice. Surface area follows the area rule, not the volume rule.

Example: A figure of area 20 is dilated by a scale factor of 4. Find the new area.
Step 1. Area is asked for, and a dilation stretches both dimensions, so the factor is used twice — the area rule.
Step 2. New area $= 20 \cdot 4^2 = 320$. Using the factor only once would give 80, which is the length rule applied to an area. $\Rightarrow$ new area $=$ **320**

Try it: A figure of area 7 is dilated by a scale factor of 5. Find the new area.

check: **175**

Skill 2 — Scaling a volume: multiply by the cube of k

Rule: a volume is built from three perpendicular lengths, so a dilation uses the factor three times: $V' = k^3 V$.

The classic slip: using k^2 on a volume (or k^3 on a surface area). Match the exponent to what is being measured — paint on the outside is an area, sand on the inside is a volume, and they scale by *different* amounts for the same k.

Example: Two similar solids have linear scale factor 3. The smaller has volume 15. Find the volume of the larger.
Step 1. A volume is asked for, and all three dimensions stretch, so the factor is used three times — the volume rule.
Step 2. Larger volume $= 15 \cdot 3^3 = 405$. Squaring instead would give 135, the area rule misapplied to a solid. $\Rightarrow$ larger volume $=$ **405**

Try it: Two similar solids have linear scale factor 4. The smaller has volume 6. Find the volume of the larger.

check: **384**

Skill 3 — Finding k from an area or volume ratio

Rule (running the rule backwards):

$$k = \sqrt{\frac{\text{area ratio}}{1}} \qquad \text{and} \qquad k = \sqrt[3]{\frac{\text{volume ratio}}{1}}$$

A ratio of areas or volumes is **never** the scale factor itself. Take the root that matches the measurement, then confirm by scaling forwards.

Example: Two similar polygons have areas in the ratio 81 to 16. Find the linear scale factor.

Step 1. The given ratio compares areas, and an area ratio is the square of the linear factor, so undo the squaring with a square root.

Step 2. $k = \sqrt{81/16} = 2.25$. Reporting the ratio itself would give about 5.06, which is a ratio of areas, not of lengths. $\Rightarrow \quad k = \mathbf{2.25}$

Try it: Two similar solids have volumes in the ratio 216 to 27. Find the linear scale factor. (Take the cube root.)

check: **2**

(!) Watch out: the exponent counts *dimensions being scaled*, not the shape of the figure. Perimeter and slant height scale by k; area, surface area and any cross-section scale by k^2; volume and capacity scale by k^3. Always finish by scaling forwards to check.